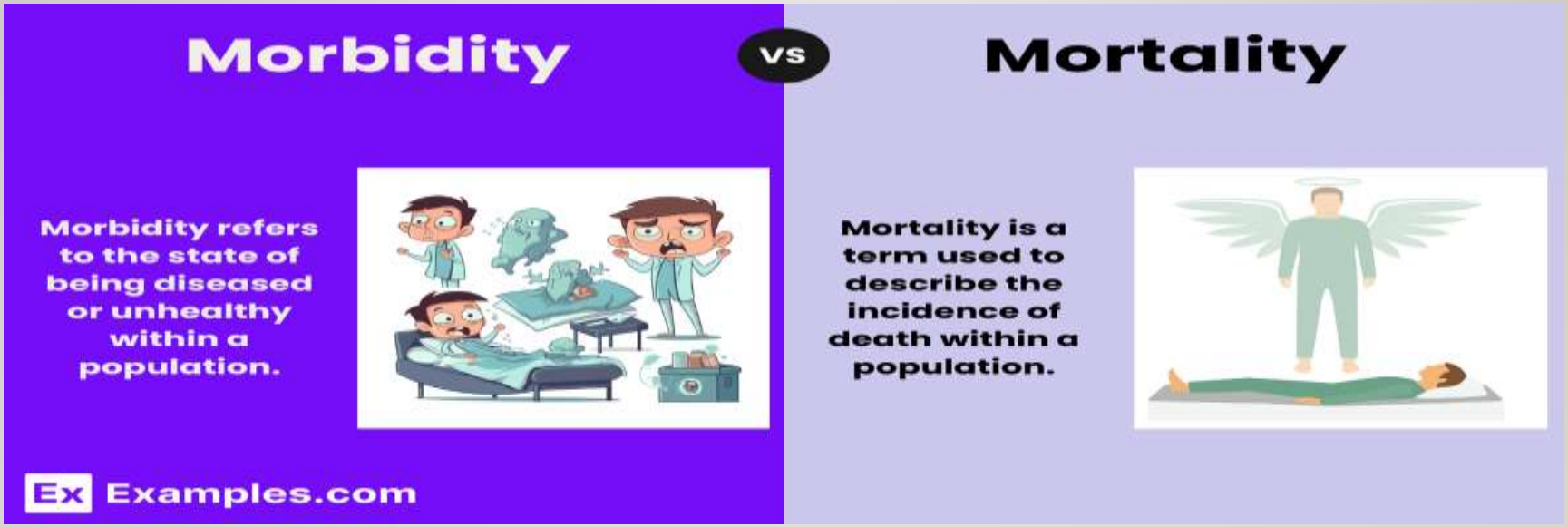
MORBIDITY,FERTILITY INDICATORS



EPIDEMIOLOGY AND DATA MANAGEMENT

DR. TASNEEM MOHAMMED ALI

LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Define morbidity, incidence, and prevalence
- Differentiate between incidence and prevalence
- Describe birth, fertility, and mortality indicators
- Apply formulas to calculate rates

INTRODUCTION

Epidemiologists describe and compare the occurrence of disease within a population.

Understanding these measurements helps identify disease determinants and health needs.

QUANTITATIVE MEASUREMENTS

- Ratio: Relationship between two independent quantities (e.g., male:female = (3:1
- Proportion: Relation of part to whole (X/Y × (100
- Rate: Probability of an event in a defined population during a specified time (X/Y × K)

MORBIDITY MEASURES

Morbidity refers to the frequency of illness in a population.

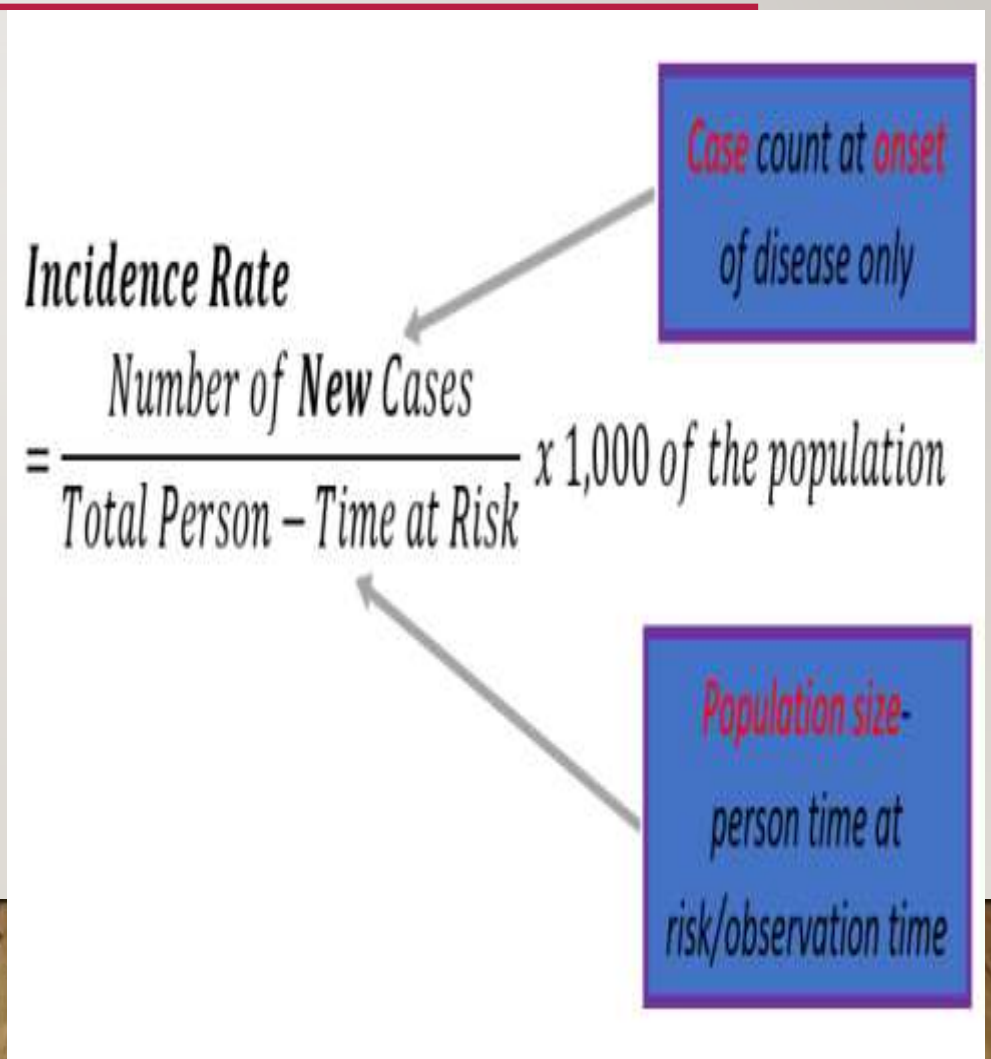
It includes:

- Incidence rate
- Prevalence rate
- Attack rate
- Secondary attack rate

INCIDENCE RATE FORMULA

- Incidence rate = (Number of new cases of disease during a given time ÷ Population at risk) × 1000

- • Indicates risk of developing disease
- • Best used for acute diseases
- • High incidence = High occurrence of disease



PREVALENCE RATE FORMULA

Prevalence rate = (Number of existing cases during a specified time ÷ Total population) × 1000

Types:

- Point prevalence (at a specific time)
- Period prevalence (over a time interval)

Prevalence = Incidence × Duration

DIFFERENCES BETWEEN INCIDENCE & PREVALENCE

- Incidence focuses on new cases and risk
- Prevalence includes all existing cases
- Incidence = Acute diseases; Prevalence = Chronic diseases
- Prevalence depends on incidence and duration of disease

Unraveling the Mystery of

PREVALENCE vs. INCIDENCE!

Ever wondered about the difference between prevalence and incidence? These two terms are often used in epidemiology and public health to measure the occurrence of diseases, but they represent different aspects of the same phenomenon.

PREVALENCE

- Represents the proportion of individuals in a population who have a specific disease at a specified point in time.
- Reflects both existing and new cases of the disease.
- Helps in understanding the overall burden of the disease in a population.

INCIDENCE!

- Refers to the rate of new cases of a disease that develop within a defined time period.
- Provides insight into the risk of developing the disease.
- Useful in studying the causes and risk factors associated with the disease.

So, next time you hear about disease stats, remember the difference

ATTACK AND SECONDARY ATTACK RATES

- Attack rate = (New cases ÷ Total at risk) × 100
- Used during outbreaks.

- Secondary attack rate = (New cases among contacts ÷ Susceptible exposed) × 100
- Measures disease spread within close groups.

FERTILITY INDICATORS

- • Crude Birth Rate (CBR)
- • General Fertility Rate (GFR)
- • Age-Specific Fertility Rate (ASFR)
- • Total Fertility Rate (TFR)
- • Gross & Net Reproduction Rates (GRR, NRR)
- • Fecundity Rate

CRUDE BIRTH RATE (CBR)

- CBR = (Number of live births ÷ Mid-year population) × 1000

- Used to evaluate natural population increase and family planning outcomes.

GENERAL & AGE-SPECIFIC FERTILITY RATES

- GFR = (Live births ÷ Women aged 15–49) × 1000
- ASFR = (Live births in age group ÷ Women in same age group) × 1000

- ASFR identifies fertility patterns across age groups.

TOTAL FERTILITY AND REPRODUCTION RATES

- • Total Fertility Rate (TFR): Average number of children a woman would bear (15–49 years)
- • Gross Reproduction Rate (GRR): Female live births per 1000 women
- • Net Reproduction Rate (NRR): Adjusted for women surviving to childbearing age

THANK YOU

MORTALITY INDICATORS

- Mortality measures death frequency in a population.
- • Crude Death Rate (CDR)
- • Age-Specific Death Rate (ASDR)
- • Infant Mortality Rate (IMR)
- • Maternal Mortality Rate (MMR)
- • Case Fatality Rate (CFR)
- • Cause-Specific Mortality Rate (CSMR)

CRUDE DEATH RATE (CDR)

- CDR = (Total deaths ÷ Mid-year population) × 1000

- Reflects general mortality but varies by age structure of population.

INFANT AND CHILD MORTALITY RATES

- IMR = (Deaths under 1 year ÷ Live births) × 1000
- • Neonatal MR: Deaths <4 weeks
- • Post-neonatal MR: Deaths 4 weeks–1 year
- • Under-5 MR: Deaths of children aged 1–4 years

PERINATAL AND MATERNAL MORTALITY

- • Perinatal MR = (Stillbirths + Early neonatal deaths ÷ Total births) × 1000
- • Maternal MR = (Maternal deaths ÷ Live births) × 100,000

- Indicators of quality of maternal and newborn care.

CASE FATALITY & CAUSE-SPECIFIC RATES

- • CFR = (Deaths from specific disease ÷ Total cases) × 100
- • CSMR = (Deaths from specific cause ÷ Population) × 100,000

- CFR indicates disease severity; CSMR shows burden of specific causes.

PROPORTIONATE MORTALITY RATIO (PMR)

- PMR = (Deaths from a given cause ÷ Total deaths) × 100

- Indicates the relative importance of diseases as causes of death.